13. Fishing to survive — rivers, lakes and sea

Oracle OS guide. Based on the FAO "Fisherman's Workbook" and FAO Training Series, US Army FM 3-05.70/21-76 (fishing chapters) and MSF/WHO (food safety) — in the library **with drawings of hooks, knots, traps and nets**. In normal times obey the law: fishing licences (England/Wales: Environment Agency rod licence; Scotland: salmon/trout permissions; Portugal: recreational fishing licence), minimum sizes and seasons. Nets, set lines and traps are usually illegal outside an emergency.

Where and when fish are

- **Rivers**: slack water behind rocks, undercut banks, shade under trees, where two streams meet, the tail of deep pools, near bridges.
- **Lakes**: weed beds, jetties, drop-offs, stream inlets; early morning and evening in the shallows, midday deeper.
- **Sea**: rocks and piers, beaches with gentle surf (fish just behind the breakers), estuaries on a rising tide, channels between sandbanks.
- Best times: **dawn and dusk**, before storms, at the turn of the tide.

Improvised gear

- **Line**: kit fishing line, dental floss, paracord inner strands, sewing thread, twisted bark fibres (see "Ropes and knots").
- **Hook**: safety pin, bent and sharpened wire, thorns (hawthorn, bramble), carved bone or hardwood, can ring-pulls.
- **Gorge hook**: 2–3 cm stick or bone sharpened at both ends, tied in the middle; the fish swallows it with the bait and it jams crosswise.
- **Bait**: worms (dig damp soil, under stones), grubs, insects, crickets, fish guts and eyes, mussels, kneaded bread, sweetcorn. Predators hit shiny moving lures (scrap of tin, feather).
- **Float**: dry stick, cork, thick feather. **Weight**: small stone or nut.

Methods

- **Handline**: line wound on a stick; cast with the wrist, feel for bites and strike firmly.
- **Improvised rod**: flexible 2–3 m pole (willow, hazel), line tied at the tip and along the pole.
- **Set lines** (emergency): several hooks on a line strung between banks or from a springy branch; check at dawn and dusk.
- **Bottle/funnel trap**: large bottle cut, neck inverted inside, bait inside, weighted; small fish and crabs get in and can't get out. Bigger version woven from willow in a cone.
- **Stone fish trap / tidal weir**: semicircle wall of stones on a beach or shallow stream with a narrow entrance; the falling tide traps fish.
- **Net** (emergency): hand net (shirt/cloth on a branch hoop) for small fish in shallows; gill nets per the FAO instructions in the library.

- **Spear**: pole with a sharpened forked tip, in shallow still water; aim below the fish you see (water bends the image).
- Night fishing with a light attracts fish and squid.

Shellfish and other water food

- **Mussels, cockles, clams**: only from clean water, away from sewage and harbours; **never during "red tides"/algal blooms** or when warnings are posted — the toxins survive cooking. Must be alive (close when tapped). Cook until open; discard those that stay shut.
- **Crabs and crayfish**: baited drop net or by hand under stones; cook thoroughly.
- **Seaweed** (sea lettuce, laver/nori, kelp): from clean water, attached to rocks (not loose on the beach); wash well.
- Avoid: pufferfish (poisonous), unfamiliar spiny or scaleless reef fish in the tropics (ciguatera), fish found dead, fish from polluted water or near outfalls.

Preparing fish

1. Kill it straight away (sharp blow to the head) — keeps better.
2. **Scale** (tail to head), **gut**: slit the belly from vent to gills, remove guts and gills, rinse.
3. Fresh fish: bright eyes, red gills, firm flesh, sea smell. Cloudy eyes, brown gills, soft flesh or strong odour → **throw away**.
4. **Always cook freshwater fish thoroughly** (parasites): until the flesh is opaque and flakes off the bone. Skewered over embers, wrapped in leaves/clay, or in soup (uses everything).
5. Guts make bait or fertiliser; bury them away from camp (they attract animals).

Preserving

- **Smoking**: thin fillets over cool hardwood smoke (oak, beech, fruit wood), 1–2 days to preserve.
- **Drying**: thin fillets in sun and wind, protected from flies, until hard.
- **Salting**: layers of fish and coarse salt; then dry. See "Food preservation".

Water safety while fishing

Unstable banks, slippery rocks and freak waves kill anglers every year. Life jacket, never alone on rocks, check the tide table, watch the sea for 10 minutes before going down. See "Water, rivers and sea".